Digital Economics Semester 8 - Work in Progress

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Project: *Hybrid ML-MCDM Framework for EV Battery End-of-Life Routing in India*
Course: Digital Economics (Semester 8)
Group 5

Group 5 Members

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1. Status Summary

1.1 Completed milestones

- **Data acquisition** — 13 datasets totaling ~113 GB downloaded, audited, and unified into a single 2.20-million-row parquet
- **Data preprocessing** — chemistry-aware filters, z-outlier flags, anti-leakage battery-level splits, full feature engineering pipeline

- **PyBaMM synthetic generation** — 70 Indian-context cells across 4 thermal profiles and 2 chemistries with verified thermal-stress ordering
- **PyBaMM validation** — 50-cycle validation passed on the SNL NMC reference cell (RMSE 1.66 %, Pearson r = 0.95)
- **Comprehensive 5-layer EDA** — 8 analysis scripts plus an executive-summary report, 34 quantitative figures
- **ML training pipeline** — 5 models trained end-to-end with checkpoints, training logs, stratified per-source and per-chemistry metrics, plus a leave-one-source-out cross-dataset generalization study
- **MCDM stack** — 6 canonical criteria fixed, 15 literature weights extracted from 8 papers, Fuzzy BWM-TOPSIS implemented and verified
- **Regulatory parsing** — BWMR (42 tables), EU Annex XIII (31 fields), GBA Battery Pass — all parsed and integrated into the unified DPP schema
- **Unified DPP schema** — JSON Schema Draft 2020-12, 9 categories reconciling EU + GBA + BWMR
- **End-to-end testing** — full pipeline (battery → SoH → grade → routing → DPP) verified on three reference cells, all schema-validated

1.2 Currently in progress

- Iteration-2 ML redesign: feature engineering with dQ/dV peak-position, Incremental Capacity, and Severson Δlog(Q(V)) features; LSTM regularization recipe; cross-source domain adaptation
- Five-scenario MCDM sensitivity sweep operationalizing the BWMR-vs-EU comparison
- Continued refinement and revision of the conceptual / methodology diagrams

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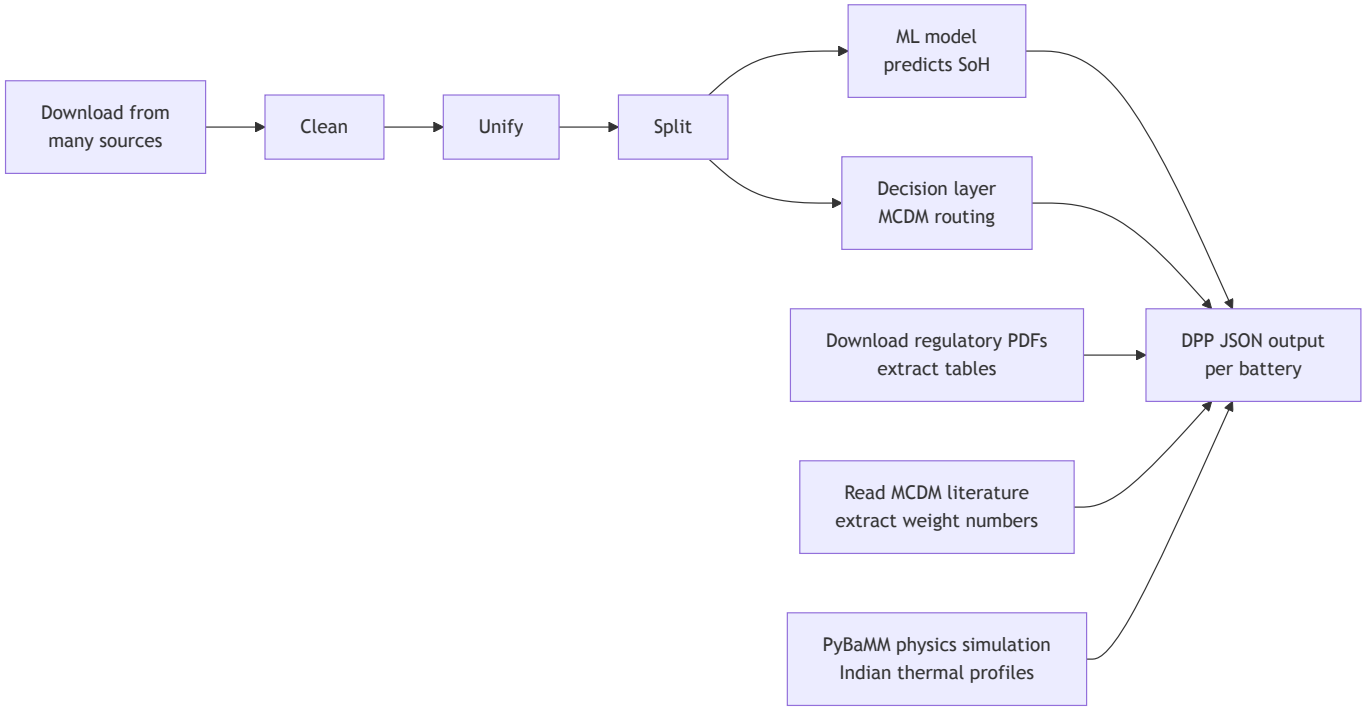
2. Project Overview

The project builds a hybrid Machine Learning + Multi-Criteria Decision-Making framework that decides what should happen to retired Electric Vehicle (EV) batteries — reuse them in stationary energy storage, refurbish modules, or send them straight to recycling — under both Indian (BWMR) and European (EU 2023/1542) regulatory constraints. End-to-end, the framework consumes battery cycling telemetry, predicts each cell's State-of-Health (SoH) and Remaining-Useful-Life (RUL), classifies it into A/B/C/D health bands, ranks four end-of-life routes via Fuzzy BWM-TOPSIS, and emits a Digital Product Passport (DPP) JSON record that satisfies both regulatory regimes.

How the project compares to a typical ML pipeline

A typical ML project follows a familiar shape: *download data → clean → split → train → evaluate → deploy*. This project follows the same shape for its core machine-learning pipeline, but adds three side-rails because the goal is not just to predict SoH — it must also make a regulatory-compliant routing decision and emit a per-battery DPP JSON record. That requires extracting fields and tables from regulatory texts (BWMR, EU Annex XIII, GBA Battery Pass), pulling decision weights from the MCDM literature, and running PyBaMM physics simulations to fill in the India-specific data gap that no public dataset covers.

The resulting flow looks like this:



The left half is the conventional ML pipeline applied to multi-source battery cycling data; the bottom three lanes are the project-specific additions that turn a SoH prediction into a regulatory-compliant routing decision and a Digital Product Passport.

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3. Literature Review

3.1 Scope

We reviewed **58 peer-reviewed papers and 2 government policy briefs**, organized across six thematic clusters that span the three literatures the project sits at the intersection of (ML, MCDM, DPP) plus the regulatory and physics-simulation context that anchors it.

3.2 Six themes covered

Theme	What it covers	Papers
T1 – Hybrid ML for SoH/RUL	XGBoost, LSTM, Transformer, CNN-BiLSTM models for State-of-Health and Remaining-Useful-Life prediction	12
T2 – MCDM for circular economy	BWM, TOPSIS, fuzzy variants for end-of-life route selection	11
T3 – Battery Digital Product Passport	EU Annex XIII, GBA Battery Pass schemas, blockchain for batteries	7
T4 – Anomaly / fault detection	IsolationForest, VAE, autoencoders for cell-level faults	6
T5 – India regulatory + circular SC	BWMR 2022 + amendments, Indian drive cycles, EPR economics	11
T6 – Physics simulation + synthetic data	PyBaMM, GAN/VAE-based augmentation	8

3.3 Five research gaps the project addresses

1. **ML × MCDM integration void** — Zero of 58 papers simultaneously cover both ML SoH/RUL prediction and MCDM ranking. Existing MCDM papers use static expert judgments; existing ML papers stop at SoH and never feed a routing decision.
2. **DPP layer divorced from operational data** — Existing DPP papers describe schemas (40+ attributes) but none expose live ML predictions or anomaly history. The most-important DPP attributes are also documented as the least available.
3. **India context is structurally absent** — Of 58 papers, only 11 engage with Indian regulation/operations, isolated within Theme 5. The Indian Driving Cycles (IDC, MIDC) defined by Indian academic labs have never been used in a PyBaMM physics simulation.
4. **Anomaly detection terminates at the alarm** — Every anomaly-detection paper stops at producing fault flags; none feed the flag into a downstream routing decision as a Safety override.
5. **Synthetic-data pipelines lack physics × generative × regional priors** — Physics-based synthesis produces generic trajectories; generative methods re-train on the same lab benchmarks. No paper combines both with India-specific drive-cycle conditioning.

The novelty of this project is **integration**: a chain (ML → grade → MCDM → dual-compliance DPP) anchored in Indian-context PyBaMM data — a combination none of the 58 reviewed papers cover.

3.4 Representative papers (10 of 58)

Authors	Year	Method	Why it matters
Jin et al.	2025	LSTM + XGBoost + Binary Firefly	Direct template for our hybrid ML stack
Xiao et al.	2024	XGBoost + TreeSHAP	SHAP explainability precedent
Park et al.	2020	LSTM RUL on multi-channel charging	Foundational LSTM-RUL paper
Song et al.	2020	XGBoost SoH	Foundational XGBoost-SoH paper
Neri et al.	2024	BWM-TOPSIS for DLT-DPP	Methodological gold standard for our MCDM stack
Ma et al.	2024	Pathway decisions for reuse/recycling	Most-cited EoL pathway paper (146 citations, *Nature Communications*)
Aishwarya et al.	2025	Fuzzy AHP-TOPSIS for EV battery SC	Indian-context circular-supply-chain MCDM
Pohlmann et al.	2024	DPP stakeholder requirements	Empirical evidence of DPP attribute unavailability
NITI Aayog	2022	India ACC battery reuse-recycling market	Government-anchored Indian market sizing
Sulzer et al.	2021	PyBaMM	The simulator we use for Indian-context cells

The full 58-row review with DOIs, key findings, verbatim quotes, and per-paper relevance is maintained as a separate working document.

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4. Datasets

We collected approximately **113 GB of raw cycling data** plus parsed regulatory corpora, organized in five layers.

4.1 Layer 1 — Real cycling data (ML training input)

#	Dataset	Size	Format	Origin
1	BatteryLife (HuggingFace)	86 GB	`.pkl` (1,382 files)	Tan et al. 2025, KDD '25
2	Stanford OSF Onori (second-life)	17 GB	`.zip` (13 archives)	Pozzato et al., Onori lab – *Data in Brief* 2022
3	Stanford OSF 8jnr5 (sister)	2.15 GB	`.zip`	Onori lab – INR21700-M50T cells
4	NASA Randomized & Recommissioned	3.2 GB	`.csv`	NASA Prognostics Center of Excellence
5	NASA PCoE (Kaggle mirror)	586 MB	`.csv`	Saha & Goebel 2007

#	Dataset	Size	Format	Origin
6	NASA Randomized v1 (Kaggle, bonus)	1.20 GB	`.mat` × 77	Bills et al. 2023
7	CALCE (Kaggle)	1.05 GB	`.xlsx` (619 files)	University of Maryland CALCE

Why this combination: BatteryLife is the largest unified public corpus and covers 18 sub-corpora across 7 chemistries. NASA and CALCE are the most-cited public benchmarks. Stanford OSF is the only public second-life dataset, essential for our retired-EV focus. NASA Randomized provides realistic non-laboratory duty cycles.

4.2 Layer 2 — Synthetic Indian-context data

We use PyBaMM (v25.4.0) with the OKane2022 + canonical degradation parameter set for NMC cells, and Prada2013 with isothermal SPMe for LFP. Generated **70 PyBaMM-synthetic cells** (50 NMC + 20 LFP), each at 4 thermal profiles (Bengaluru-mild, Mumbai-monsoon, Delhi-summer, Rajasthan-extreme).

4.3 Layer 3 — Regulatory corpora

- **BWMR 2022 + 4 amendments** (Indian battery waste regulation, parsed to English text and 42 extracted tables)
- **EU Regulation 2023/1542 + Annex XIII** (31 mandated DPP fields parsed across 2 access tiers)
- **GBA Battery Pass v1.2.0** (DIN DKE SPEC 99100:2025-02 — used as the structural base for our schema)

4.4 Layer 4 — India market data

CPCB EPR procurement table (9 metals: target vs procured vs available); ICEA-Accenture growth projections (19 candidates including 200–250 kT lithium-ion battery material by 2030); WRI "Second Charge" market report.

4.5 Layer 5 — Literature MCDM weights

15 numerical weights extracted from 8 papers in the MCDM theme, mapped to 6 canonical criteria (SoH, Value, Carbon, Compliance, Safety, EPR Return), aggregated to triangular fuzzy numbers (TFNs) and normalized for Fuzzy BWM input.

4.6 Common variables across multi-source corpus

The seven raw cycling datasets each had idiosyncratic schemas, so we built per-source loaders that emit a single unified 32-column schema. The unified schema defines a common set of variables present for every battery regardless of source — identity (battery ID, cycle number,

source, chemistry, form factor, nominal capacity, second-life flag), per-cycle electrical measurements (capacity, voltage statistics, current statistics, temperature statistics, charge/discharge times, internal resistance, coulombic efficiency), z-score quality flags, and per-battery time-series derivatives (capacity deltas, rolling means and standard deviations, cumulative cycle count). This unified schema is the foundation of every downstream step.

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5. Variables and Feature Selection

5.1 The 17 numeric features used by the ML models

After exploratory data analysis, five sparse features (internal resistance and four temperature columns) were excluded from the default feature set; the remaining 17 numeric features feed the models:

- Capacity-related: `nominal_Ah`, `capacity_Ah`, `capacity_delta`, `capacity_roll5_mean`, `capacity_roll5_std`, `capacity_roll20_mean`, `capacity_roll20_std`
- Voltage statistics: `v_min`, `v_max`, `v_mean`
- Current statistics: `i_min`, `i_max`, `i_mean`
- Duty-cycle: `charge_time_s`, `discharge_time_s`
- Efficiency: `coulombic_eff`
- Calendar life: `cycle_count_so_far`

Plus categorical one-hot expansions (16 columns total) for chemistry (7 levels), form factor (5 levels), and source family (4–5 levels). **Final design matrix: 33 features.**

5.2 Targets

- **State of Health (SoH%)** — capacity divided by nominal capacity, expressed as a percentage in the range [0, 150 %]
- **Remaining Useful Life (RUL)** — cycles remaining until SoH drops below 80 %, right-censored at the maximum cycle observed for each cell

5.3 Battery health grade thresholds

Grade	SoH range	Recommended end-of-life route
A	SoH > 80 %	Grid-scale ESS (second-life)
B	60 % < SoH ≤ 80 %	Home / Distributed ESS
C	40 % < SoH ≤ 60 %	Component / module reuse
D	SoH ≤ 40 %	Direct recycling

Thresholds anchored in Severson 2019 and the EU 2023/1542 80 % retirement criterion.

5.4 The 6 MCDM canonical criteria

#	Criterion	Type	Papers with weight	Normalized weight
C1	SoH	benefit	0 (sensitivity-only)	–
C2	Value	benefit	1	0.360
C3	Carbon	cost	3	0.166
C4	Compliance	benefit	6	0.266
C5	Safety	cost	1	0.071
C6	EPR Return	benefit	4	0.137

Compliance has the strongest evidence base. The four MCDM alternatives ranked by TOPSIS are: Grid-scale ESS, Home/Distributed ESS, Component Reuse, Direct Recycling.

5.5 The 9 DPP schema categories

The unified DPP schema reconciles EU Annex XIII §1–4, GBA Battery Pass v1.2.0, and India BWMR 2022/2024/2025 EPR schedules into nine categories: identity; chemistry and composition; performance and durability; state of health; carbon footprint; supply-chain due diligence; circularity and end-of-life; labels and compliance; dismantling and safety. The `circularity_and_eol.epr_compliance` block is the BWMR-specific carve-out.

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6. Data Preprocessing

6.1 The 5-step pipeline

1. **Per-source loaders** — read each raw dataset and emit standardized per-cycle records, normalizing column names and units.
2. **Unifier** — concatenate all sources into a single parquet file, apply a chemistry-aware voltage filter (Zn-ion 0.8–2 V, Li-ion 2–4.5 V, Na-ion 1.5–4.5 V), and compute z-score outlier flags.
3. **Splits** — battery-level 70/15/15 split (1,084 / 217 / 220 cells), stratified by source × chemistry × second-life flag (29 strata), with anti-leakage check ensuring zero battery-ID overlap. Synthetic cells are pinned to the train split.
4. **Feature engineering** — compute SoH%, compute RUL, fit the StandardScaler on the train split only, one-hot encode categorical context, drop a handful of label-quality issues identified during EDA.
5. **Stage-EDA audit** — eight self-contained EDA scripts producing 34 quantitative figures plus an executive-summary report.

6.2 Final preprocessed corpus

- **Unified ML input**: 228.7 MB parquet, **2,204,054 rows × 32 columns**, **1,521 batteries**, **34 source codes**, **7 chemistries** (NMC, LFP, NCA, LCO, LMO, Zn-ion, Na-ion)
- **Splits**: train 1,084 batteries, val 217, test 220 (anti-leakage verified, KS statistic on SoH train-vs-test < 0.10)

6.3 EDA highlights

Eight focused EDA analyses surfaced meaningful structure in the corpus:

- **Distribution audit:** NMC dominates 75 % of rows; the corpus skews late-life (mean SoH 0.35), giving the framework rich exposure to grade-C/D cells which most prior work lacks.
- **Knee-point analysis:** approximately 80 % of detected knee points fall after the 50 % mark of cycle life, validating our 30-cycle LSTM context window choice.
- **Anti-leakage:** zero battery-ID overlap across train/val/test; chemistry composition preserved within ~5 percentage points across splits.
- **Synthetic data validation:** PyBaMM-generated NMC cells show the correct thermal-stress ordering of fade rates (Bengaluru < Mumbai < Delhi < Rajasthan), confirming PyBaMM thermal coupling is wired correctly.
- **Regulatory landscape:** BWMR EV recovery target ramps from 70 % to 90 % over 2024–27; CPCB national procurement averages only 58 % of target across metals — empirical motivation for high MCDM weight on Compliance and EPR Return.
- **MCDM input quality:** strict evidence threshold (≥3 papers) covers Compliance, EPR Return, and Carbon; the remaining three criteria are handled via a five-scenario sensitivity sweep.

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7. Initial Modelling and Framework

7.1 Five models trained end-to-end

Model	Purpose	Architecture	Training time
XGBoost SoH regressor	Predict State-of-Health (%)	Depth 9, 996 trees (Optuna-tuned, 50 trials)	14.65 s
LSTM RUL forecaster	Predict Remaining-Useful-Life (cycles)	2 layers × hidden 128, dropout 0.2, sequence length 30	49.5 min
VAE anomaly detector	Flag unusual cycle records	Encoder/decoder [128, 64, 32], latent dim 12	30 min
IsolationForest	Tree-based anomaly baseline	200 trees, contamination 0.05	2.78 s
SoH → Grade classifier	Map SoH to A/B/C/D + EoL route	Threshold rule on XGBoost SoH	derived

Total wall-clock for the orchestrated full run: 86 minutes. All models save checkpoints, training logs, and per-run metrics; a comprehensive set of evaluation figures (loss curves, predicted-vs-actual, residual diagnostics, confusion matrices, feature importance) is generated automatically.

7.2 Preliminary test-set metrics

Model	Metric	Result	Target
XGBoost SoH	R^2	0.9991	> 0.95
XGBoost SoH	RMSE (% SoH)	1.094 %	< 2 %
XGBoost SoH	MAE (% SoH)	0.402 %	< 1.5 %
LSTM RUL	R^2	0.891	–
LSTM RUL	RMSE (% of range)	4.54 %	< 2 %
Grade classifier	Accuracy	0.9853	≥ 0.90
Grade classifier	F1 macro	0.9727	≥ 0.85
VAE anomaly	Test flag rate	5.07 %	≈ 5 %
IsolationForest	Test flag rate	5.38 %	≈ 5 %

These are first-iteration results; further improvements to architectures, regularization, and feature engineering are in progress.

7.3 Integration story — single-battery flow

A trained battery flows through the framework as follows: cycle row → XGBoost predicts SoH → grade-from-SoH thresholding → grade-specific Fuzzy BWM-TOPSIS routing → rank-1 alternative chosen → DPP JSON assembled and schema-validated.

This pipeline is verified end-to-end on three reference cells:

Battery	Cycles	SoH	Grade	Recommended route	DPP coverage
Zn-ion coin cell	391	50.5 %	C	Direct Recycling (closeness 0.85)	47.7 %
SNL 18650 NMC reference	1,321	73.8 %	B	Home/Distributed ESS (closeness 0.73)	48.7 %
Synthetic Delhi NMC	100	98.9 %	A	Grid-scale ESS (closeness 0.78)	49.9 %

All three pass JSON-Schema validation. Carbon-footprint and supply-chain blocks remain placeholders pending downstream LCA integration.

7.4 Stratified evaluation — per chemistry

Headline metrics can hide chemistry-specific behaviour, so every test-set metric is computed per chemistry. For SoH regression on the test set:

Chemistry	Test rows	R^2
NMC	209,743	0.9997
LFP	54,924	0.998
NCA	8,815	0.994
LCO	6,896	0.986
Zn-ion	14,161	0.957
Na-ion	2,009	0.944

For the grade classifier, F1-macro per chemistry ranges from 0.474 (Na-ion, 2,009 rows) to 0.982 (NMC, 209,743 rows), reflecting variation in cycle-range coverage across chemistries. Improvements for under-represented chemistries are in progress.

7.5 Cross-source generalization study (Leave-One-Source-Out)

To assess genuine cross-dataset transfer, we trained XGBoost on nine sources at a time and evaluated on the held-out tenth. Strongest results from the study:

Held-out source	Test rows	LOSO R^2
Stanford	39,329	0.999
Stanford 2	188,621	0.997
MATR	139,997	0.809
Tongji	59,038	0.692

Median LOSO R^2 across the 10-source study is 0.348, confirming that Stanford-protocol Li-ion 18650 cells transfer cleanly across labs while idiosyncratic protocols and chemistry-orphan sources require additional treatment. Improvements in feature engineering and domain-adaptation techniques are in progress for the under-represented sources.

7.6 PyBaMM physics validation

The PyBaMM OKane2022 + canonical degradation stack (DFN with SEI, partially-reversible lithium plating, stress-driven loss-of-active-material, SEI on cracks, particle mechanics, lumped thermal) was validated against the SNL 18650 NMC reference cell. The 50-cycle validation passes (RMSE 1.66 %, Pearson r = 0.95). Longer-horizon shape match is preserved (Pearson r = 0.93 over 500 cycles); magnitude calibration to a different cell chemistry is documented and

the synthetic cells are used for training augmentation, with the calibration improvements scheduled for the next iteration.

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8. MCDM and DPP Framework

8.1 Fuzzy BWM-TOPSIS routing

Criterion weights are derived from the literature using Fuzzy BWM (no expert surveys needed). The 15 weights extracted from 8 MCDM papers are aggregated as triangular fuzzy numbers per criterion, defuzzified via centroid, and normalized. The 4 alternatives (Grid-scale ESS, Home/Distributed ESS, Component Reuse, Direct Recycling) are then ranked by TOPSIS using grade-specific decision matrices.

8.2 RQ2 working hypothesis (BWMR vs EU regulatory weighting)

Under BWMR-weighted scenarios (high weight on Compliance and EPR Return), we expect Direct Recycling and authorized refurbishment routes to rank highest, particularly for Grades C and D — driven by India's BWMR recovery-target schedule (70 → 80 → 90 % by FY 2026-27) and the documented 58 % CPCB national procurement gap. Under EU-weighted scenarios (high weight on Carbon footprint and Compliance), we expect Grid-scale ESS to rank higher for Grades A and B because carbon-footprint avoidance from extending first-use is largest there, while Direct Recycling becomes preferred only at Grades C/D where capacity-fade economics dominate. Stability is reported via Spearman ρ on the 4-alternative ranking across five sensitivity scenarios. Falsification criterion: if mean Spearman $\rho < 0.20$ between BWMR- and EU-weighted rankings across all grades, the hypothesis is rejected and the conclusion becomes that current regulatory weights produce convergent recommendations — itself a policy-relevant finding.

8.3 Digital Product Passport

Each DPP JSON is validated against a single unified JSON Schema (Draft 2020-12) that simultaneously satisfies EU Annex XIII fields (with access tiers), GBA Battery Pass v1.2.0 structure, and BWMR EPR fields. Three DPPs have been generated end-to-end and validated; coverage on the populated categories ranges 47.7–49.9 %. Carbon-footprint and supply-chain due-diligence blocks intentionally remain marked as placeholders pending LCA and sourcing data integration.

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9. Methodology Diagrams and Figures

9.1 Quantitative figures produced

Quantitative figures are generated automatically as a by-product of the EDA and training pipelines. They include per-column missingness charts, source × chemistry coverage matrices,

chemistry violins, voltage range boxplots, capacity-vs-cycle trajectories, fade-rate distributions, knee-point histograms, correlation heatmaps, outlier flag counts, per-split distribution comparisons, synthetic-vs-real reference plots, BWMR target trajectories, CPCB procurement bars, the MCDM coverage matrix, the TFN range plot, plus per-model training loss curves, predicted-vs-actual scatter plots, residual diagnostics, confusion matrices, feature importance bars, anomaly score distributions, and the cross-source generalization bar chart. All figures use a consistent paper-ready theme (300 dpi PNG plus PDF), saved automatically by the training and EDA scripts.

9.2 Conceptual / methodology diagrams

Conceptual and methodology diagrams have been prepared, covering high-level architecture, end-to-end pipeline, data acquisition map, processing pipeline, schema reconciliation, ML pipeline, individual model architectures, cross-source generalization methodology, Fuzzy BWM workflow, TOPSIS workflow, sensitivity analysis, DPP schema, DPP generation sequence, research methodology overview, iteration methodology, sub-team responsibilities, baseline comparison, and the end-to-end smoke-test sequence. Diagrams are produced in Mermaid, Excalidraw, and matplotlib as appropriate to each subject.

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